

CST434	NETWORK SECURITY PROTOCOLS	CATEGORY	L	T	P	CREDIT	YEAR OF INTRODUCTION
		PEC	2	1	0	3	2019

Preamble: This course helps the learners to explore various network and system security protocols. This course covers authentication protocols, firewalls and security protocols from different layers such as data link, network, transport and application. The concepts covered in this course enable the learners in effective use of security protocols for securing network applications.

Prerequisite: A fundamental knowledge in the concepts of Security in Computing.

Course Outcomes: After the completion of the course, the student will be able to

CO1	Explain authentication protocols, X.509 authentication service and Public Key Infrastructure (PKI). (Cognitive Knowledge Level: Understand)
CO2	Identify the security mechanisms in E mail security services. (Cognitive Knowledge Level: Understand)
CO3	Summarize the network and transport layer security services provided in a secure communication scenario. (Cognitive Knowledge Level: Apply)
CO4	Describe real time communication security and application layer security protocols. (Cognitive Knowledge Level: Apply)
CO5	Explain the concepts of firewalls and wireless network security. (Cognitive Knowledge Level: Understand)

Mapping of course outcomes with program outcomes

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1												
CO2												
CO3												
CO4												
CO5												

Abstract POs defined by National Board of Accreditation			
PO#	Broad PO	PO#	Broad PO
PO1	Engineering Knowledge	PO7	Environment and Sustainability
PO2	Problem Analysis	PO8	Ethics
PO3	Design/Development of solutions	PO9	Individual and team work
PO4	Conduct investigations of complex problems	PO10	Communication
PO5	Modern tool usage	PO11	Project Management and Finance
PO6	The Engineer and Society	PO12	Life long learning

Assessment Pattern

Bloom's Category	Test 1 (%)	Test 2 (%)	End Semester Examination (%)
Remember	20	20	20
Understand	50	50	50
Apply	30	30	30
Analyse			
Evaluate			
Create			

Mark Distribution

Total Marks	CIE Marks	ESE Marks	ESE Duration
150	50	100	3

Continuous Internal Evaluation Pattern:

Attendance : 10 marks

Continuous Assessment Test : 25 marks

Continuous Assessment Assignment : 15 marks

Internal Examination Pattern:

Each of the two internal examinations has to be conducted out of 50 marks. First series test shall be preferably conducted after completing the first half of the syllabus and the second series test shall be preferably conducted after completing remaining part of the syllabus. There will be two parts: Part A and Part B. Part A contains 5 questions (preferably, 2 questions each from the completed modules and 1 question from the partly completed module), having 3 marks for each question adding up to 15 marks for part A. Students should answer all questions from Part A. Part B contains 7 questions (preferably, 3 questions each from the completed modules and 1 question from the partly completed module), each with 7 marks. Out of the 7 questions, a student should answer any 5.

End Semester Examination Pattern:

There will be two parts; Part A and Part B. Part A contains 10 questions with 2 questions from each module, having 3 marks for each question. Students should answer all questions. Part B contains 2 questions from each module of which a student should answer any one. Each question can have maximum 2 sub-divisions and carries 14 marks.

Syllabus**Module-1 (Authentication Protocols)**

Authentication Protocols – Mutual authentication, One way authentication. Kerberos – Kerberos Version 4, Kerberos Version 5. X.509 Authentication service. Public Key Infrastructure (PKI) – Trust models, Revocation.

Module-2 (E-mail Security)

Pretty Good Privacy (PGP) – Operational Description, Cryptographic keys and key rings, Message format, PGP message generation, PGP message reception, Public key management. S/MIME – Functionality, Messages, Certificate processing, Enhanced security services.

Module-3 (Network Layer Security and Web Security)

Internet Protocol Security (IPSec) – Overview, IP security architecture, Authentication Header (AH), Encapsulating Security Payload (ESP), Combining Security Associations, Key management. Internet Key Exchange (IKE) - Phases. Web Security – Web security considerations. Secure Socket Layer and Transport Layer Security (SSL/TLS) – SSL Architecture, SSL protocols, Cryptographic computations, Transport layer security.

Module-4 (Real-time Security and Application Layer Security)

Real-time communication security – Perfect Forward Secrecy (PFS), Denial-of-Service protection, Endpoint identifier hiding, Live partner reassurance. Hypertext Transfer Protocol Secure (HTTPS) – Connection initiation, Closure. Secure Shell (SSH) – Transport layer protocol, User authentication protocol, Connection protocol. Secure Electronic Transaction (SET) – Overview, Features, Participants, Dual signature, Payment processing.

Module-5 (System Security and Wireless Security)

Firewalls – Firewall characteristics, Types of Firewalls, Firewall configurations, Encrypted Tunnels, Trusted systems – Data access control, The concept of Trusted Systems, Trojan horse defense. IEEE 802.11i wireless LAN security - Services, Phases of operation, Wired Equivalent Privacy (WEP), Wi-Fi Protected Access (WPA), WPA2.

Text Books

1. William Stallings, Cryptography and Network Security Principles and Practice, 4/e, Pearson Ed.
2. C. Kaufman, R. Perlman and M. Speciner, “Network Security: Private Communication in a Public World”, 2/e, PHI.

References

1. Behrouz A. Forouzan, Debdeep Mukhopadhyay, “Cryptography and Network Security”, 3/e, Tata McGraw Hill.
2. Tyler Wrightson, “Wireless Network Security A Beginner’s Guide”, 2012, Tata McGraw Hill.
3. William Stallings, “Network Security Essentials: Applications and Standards”, 4/e, Prentice Hall.
4. Schiller J., Mobile Communications, 2/e, Pearson Education.
5. Roberta Bragg et. al., “Network Security: The Complete Reference”, Tata McGraw Hill

Sample Course Level Assessment Questions**Course Outcome 1 (CO1):**

1. Identify the threats associated with user authentication over a network or Internet.
2. In the context of Kerberos, mention the significance of a realm.

Course Outcome 2 (CO2):

1. Mention the use of R64 conversion for an e-mail application.
2. Give the general structure of Private and Public Key rings in PGP.

Course Outcome 3 (CO3):

1. In AH protocol, identify the fields in an IP header which are included in MAC calculation. For each of the fields in the IP header, indicate whether the field is immutable, mutable but predictable, or mutable. Justify your decision for each field.

- Is it possible for the receiver to reorder SSL record blocks that arrive out of order?
If so, explain how it can be done. If not, why?

Course Outcome 4 (CO4):

- Devise a protocol based on a pre-shared secret key that hides identities and gives Perfect Forward Secrecy (PFS) for identity hiding. Make two variants, one in which an active attacker can learn only the initiator's identity, and one in which an active attacker can learn only the target's identity.
- Explain the tasks performed by the payment gateway during Payment Authorization in SET.

Course Outcome 5 (CO5):

- List the weaknesses of a packet-filtering router.
- Give the relevance of pair wise keys and group keys in IEEE 802.11i.
- State the design goals of firewalls.

Model Question Paper**QP CODE:****PAGES:** ____**Reg No:** _____**Name:** _____**APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY****EIGHTH SEMESTER B.TECH DEGREE EXAMINATION, MONTH & YEAR****Course Code: CST434****Course Name: NETWORK SECURITY PROTOCOLS****Max Marks: 100****Duration: 3 Hours****PART A****(Answer All Questions. Each question carries 3 marks)**

- List any three requirements of Kerberos.
- Specify the significance of key pair recovery. When is the key pair updated?
- Why does PGP generate signature before applying compression?
- List the four principal services provided by S/MIME.
- Explain the significance of Alert protocol in SSL and list out any three Alert messages with their uses.
- Specify the purpose of MAC during the change cipher spec TLS exchange.

7. What is the advantage, if any, of not including the MAC in the scope of packet encryption in SSH packets?
8. Give the significance of dual signature in SET.
9. List the IEEE 802.11i services.
10. How is the concept of association related to that of mobility in wireless networks? (10x3=30)

Part B

(Answer any one question from each module. Each question carries 14 Marks)

11. (a) Describe the requirements for a public-key certificate scheme. (8)
- (b) Explain the significance of chain of certificates. (6)

OR

12. (a) Specify the purpose of the X.509 standard. How is an X.509 certificate revoked? (8)
- (b) Describe the management functions of a PKI. What is a cross certificate? (6)
13. (a) List the services provided by PGP and explain how authentication and confidentiality are provided. (8)
- (b) Explain the functionalities provided by S/MIME. (6)

OR

14. (a) Give the format of a PGP message and specify the significance of each field in the message. (8)
- (b) Explain the enhanced security services provided in S/MIME. (6)
15. (a) Explain the parameters that identify an SSL session state. (8)
- (b) Differentiate between transport mode and tunnel mode in IPSec. (6)

OR

16. (a) The IPsec architecture document states that when two transport mode SAs are bundled to allow both AH and ESP protocols on the same end-to-end flow, only one ordering of security protocols seems appropriate: performing the ESP protocol before performing the AH protocol. Why is this approach

recommended rather than authentication before encryption?

(b) List and explain the purpose each Alert Codes supported by SSL. (6)

17. (a) Illustrate the significance of perfect forward secrecy. (6)

(b) Explain the key features provided by SET. (8)

OR

18. (a) List and explain the SSH protocols. (8)

(b) “The HTTPS capability is built into all modern web browsers”. Justify. (6)

19. (a) Explain the phases of operations in IEEE 802.11i. (8)

(b) Give the significances of Encrypted Tunnels (6)

OR

20. (a) Compare the features of three types of firewalls. (8)

(b) Compare the Wireless LAN protocols WEP, WPA and WPA2 (6)

TEACHING PLAN

No	Contents	No. of Lecture Hours (35 Hrs)
Module-1 (Authentication Protocols)(7hrs)		
1.1	Authentication Protocols – Mutual authentication, One way authentication	1
1.2	Kerberos –Version 4	1
1.4	Differences between Kerberos Version 4 and Version 5, Kerberos Version 5	1
1.5	X.509 Authentication service – Certificates, Authentication Procedures, X.509 Version 3	1
1.6	Public Key Infrastructure (PKI) – Trust models	1
1.7	Public Key Infrastructure (PKI) – Revocation	1

Module-2 (E-mail Security) (6 hrs)		
2.1	Pretty Good Privacy (PGP) – Operational Description	1
2.2	Cryptographic keys and key rings, Message format	1
2.3	PGP message generation, PGP message reception	1
2.4	PGP -Public key management	1
2.5	S/MIME – Overview of MIME, Functionality, Messages	1
2.6	S/MIME - Certificate processing, Enhanced security services	1
Module-3 (Network Layer Security and Web Security)(8 hrs)		
3.1	Internet Protocol Security (IPSec) – Overview, IP security architecture	1
3.2	Authentication Header (AH)	1
3.3	Encapsulating Security Payload (ESP)	1
3.4	Combining Security Associations, Key management	1
3.5	Internet Key Exchange (IKE) – Phases	1
3.6	Web Security – Web security considerations. Secure Socket Layer and Transport Layer Security (SSL/TLS) – SSL Architecture	1
3.7	SSL Protocols - Record Protocol, Change Cipher Spec Protocol, Alert Protocol	1
3.8	SSL Handshake Protocol, Cryptographic computations, Transport Layer Security	1
Module-4 (Real-time Security and Application Layer Security) (8hrs)		
4.1	Real-time communication security – Perfect Forward Secrecy (PFS)	1
4.2	Denial-of-Service protection, Endpoint identifier hiding, Live partner reassurance	1
4.3	Hypertext Transfer Protocol Secure (HTTPS) – Connection initiation, Closure	1
4.4	Secure Shell (SSH) – Transport layer protocol	1
4.5	User authentication protocol	1

4.6	Connection protocol	1
4.7	Secure Electronic Transaction (SET) – Overview, Features, Participants	1
4.8	Dual signature, Payment processing	1
Module-5 (System Security and Wireless Security) (6 hrs)		
5.1	Firewalls – Firewall characteristics, Types of Firewalls	1
5.2	Firewalls – Firewall configurations, Encrypted Tunnels	1
5.3	Trusted systems – Data Access Control, The Concept of Trusted Systems, Trojan Horse Defense	1
5.4	IEEE 802.11i wireless LAN security - Services, Phases of operation	1
5.5	Wired Equivalent Privacy (WEP)	1
5.6	Wi-Fi Protected Access (WPA), WPA2	1

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